Warm-up

(1) Simplify the expression $4^{7\log_4(8)-\log_4(9)}$.

(2) Find $\dfrac{d}{dx}3^{2x^2}$.

(3) Give an exact solution for the equation $6^{x-5} = 4$.

Medium

(1) Showing your work, compute $\log_3(90) + \log_3(3/2) - \log_3(5)$.

(2) Find an equation for the tangent line to the curve $y = \ln(2x^2 - 3x - 1)$ at $(2, 0)$.

(3) Compute $\dfrac{d}{dt}2^t \log_5(t)$.

(4) Compute $\dfrac{d}{dx}\left(\sqrt{x+1}\right)^x$.

(5) Compute $\dfrac{d}{dx}(5 + 9x^2)^{\ln(x)}$.

Stretch

(1) Find any maxima and minima of the function $\ln(x)/x$, and sketch the graph. Using this, determine which number is larger: e^π or π^e?

(2) Define the **hyperbolic trig functions** by

$$\sinh(x) = \frac{e^x - e^{-x}}{2}, \quad \cosh(x) = \frac{e^x + e^{-x}}{2}, \quad \text{and} \quad \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

Show that the following identities hold:

(a) $\cosh^2(x) - \sinh^2(x) = 1$
(b) $\frac{d}{dx}\sinh(x) = \cosh(x)$
(c) $\frac{d}{dx}\cosh(x) = \sinh(x)$
(d) $\frac{d}{dx}\tanh(x) = 1/\cosh^2(x)$.